

■ 3.7.3 Scalars, Vectors and Matrices

Mathematica represents matrices and vectors using lists. Anything that is not a list *Mathematica* considers as a scalar.

A vector in *Mathematica* consists of a list of scalars. A matrix consists of a list of vectors, representing each of its rows. In order to be a valid matrix, all the rows must be the same length, so that the elements of the matrix effectively form a rectangular array.

<code>VectorQ[expr]</code>	give True if <i>expr</i> has the form of a vector, and False otherwise
<code>MatrixQ[expr]</code>	give True if <i>expr</i> has the form of a matrix, and False otherwise
<code>Dimensions[expr]</code>	a list of the dimensions of a vector or matrix

Functions for testing the structure of vectors and matrices.

The list {a, b, c} has the form of a vector.

```
In[1]:= VectorQ[ {a, b, c} ]
Out[1]= True
```

Anything that is not manifestly a list is treated as a scalar, so applying `VectorQ` gives **False**.

```
In[2]:= VectorQ[ x + y ]
Out[2]= False
```

This is a 3×2 matrix.

```
In[3]:= Dimensions[ {{a, b, c}, {ap, bp, cp}} ]
Out[3]= {2, 3}
```

For a vector, `Dimensions` gives a list with a single element equal to the result from `Length`.

```
In[4]:= Dimensions[ {a, b, c} ]
Out[4]= {3}
```

This object does not count as a matrix because its rows are of different lengths.

```
In[5]:= MatrixQ[ {{a, b, c}, {ap, bp}} ]
Out[5]= False
```